

Superoptimal Block-Circulant Preconditioner for Quasi-Block-Toeplitz Harmonic Balance Systems

Detailed derivation of $\min_C \|I - C^{-1}A\|_F^2$ following Tyrtysnikov (1992), applied to harmonic balance

1. Problem setup

Consider a time-periodic linear operator with Fourier coefficients M_j for $j = -n_p, \dots, n_p$, each of size $N \times N$. Harmonic balancing with n_p perturbation harmonics ($n_p \geq n_f$) yields a block system $Ax = f$ of total size $nN \times nN$, where $n = 2n_p + 1$. The (k, j) block of A is:

$$A_{kj} = (s + ik\omega) I_N - M_0, \quad k = j \quad (1a)$$

$$A_{kj} = -M_{k-j}, \quad 0 < |k - j| \leq n_f \quad (1b)$$

$$A_{kj} = 0, \quad |k - j| > n_f \quad (1c)$$

where $k, j = -n_p, \dots, n_p$ and s is a (possibly zero) Laplace-domain shift. The diagonal blocks carry the frequency-dependent shift $ik\omega$, which breaks the block-Toeplitz structure. We write $A = D + S$, where:

$$D = \text{blkdiag}(i(-n_p)\omega I, \dots, 0, \dots, i n_p \omega I) \quad (2)$$

is block-diagonal and S is the pure block-Toeplitz part with blocks $S_{kj} = (sI - M_0)\delta_{kj} - M_{k-j}(1 - \delta_{kj})$.

2. Block-circulant structure and the DFT

A block-circulant matrix C has (k, j) block depending only on $(k - j) \bmod n$. Any such matrix is diagonalized by the block DFT:

$$C = (F_n \otimes I_N) \Lambda_C (F_n^* \otimes I_N) \quad (3)$$

where F_n is the $n \times n$ unitary DFT matrix with entries $(F_n)_{pq} = \omega_n^{pq/n}$, $\omega_n = e^{-2\pi i/n}$, and $\Lambda_C = \text{blkdiag}(\Lambda_0, \dots, \Lambda_{n-1})$ with each Λ_k an $N \times N$ matrix.

The inverse is $C^{-1} = (F \otimes I) \Lambda_C^{-1} (F^* \otimes I)$, where each diagonal block is inverted independently.

3. The optimization problem

We seek the block-circulant C that minimizes the Frobenius norm of the preconditioned residual from identity:

$$\min_{C \text{ block-circ.}} \|I - C^{-1}A\|_F^2 \quad (4)$$

Define the DFT-domain representation of A :

$$G = (F^* \otimes I) A (F \otimes I) \quad (5)$$

which is an $nN \times nN$ matrix partitioned into $N \times N$ blocks G_{kj} . Note that G is generally dense (i.e., $G_{kj} \neq 0$ for $k \neq j$) because A is only quasi-block-Toeplitz, not block-circulant. If A were block-circulant, G would be block-diagonal.

Since $F \otimes I$ is unitary, the Frobenius norm is preserved:

$$\|I - C^{-1}A\|_F^2 = \|I - \Lambda_C^{-1}G\|_F^2 \quad (6)$$

3.1. Expanding the norm

Partition the right-hand side of (6) into $N \times N$ blocks. Since Λ_C is block-diagonal, the (k, j) block of $\Lambda_C^{-1}G$ is $\Lambda_k^{-1}G_{kj}$. Therefore:

$$\|I - \Lambda_C^{-1}G\|_F^2 = \sum_{k,j} \|\delta_{kj}I - \Lambda_k^{-1}G_{kj}\|_F^2 \quad (7)$$

Separating diagonal ($k = j$) and off-diagonal ($k \neq j$) terms:

$$= \sum_k \|I - \Lambda_k^{-1}G_{kk}\|_F^2 + \sum_k \sum_{j \neq k} \|\Lambda_k^{-1}G_{kj}\|_F^2 \quad (8)$$

Each Λ_k appears only in the k th group of terms (the k th block-row of G), so the minimization over $\Lambda_0, \dots, \Lambda_{n-1}$ decouples into n independent problems. For each k , we minimize:

$$f(\Lambda_k) = \|I - \Lambda_k^{-1}G_{kk}\|_F^2 + \sum_{j \neq k} \|\Lambda_k^{-1}G_{kj}\|_F^2 \quad (9)$$

4. Solving the block optimization problem

4.1. Reformulation

Let $X_k = \Lambda_k^{-1}$. Then (9) becomes:

$$f(X_k) = \|I - X_k G_{kk}\|_F^2 + \sum_{j \neq k} \|X_k G_{kj}\|_F^2 \quad (10)$$

Using the identity $\|AB\|_F^2 = \text{tr}(ABB^*A^*)$, expand each term:

$$\|I - X_k G_{kk}\|_F^2 = \text{tr}(I) - \text{tr}(X_k G_{kk}) - \text{tr}(G_{kk}^* X_k^*) + \text{tr}(X_k G_{kk} G_{kk}^* X_k^*) \quad (11)$$

$$\sum_{j \neq k} \|X_k G_{kj}\|_F^2 = \sum_{j \neq k} \text{tr}(X_k G_{kj} G_{kj}^* X_k^*) \quad (12)$$

Combining, and defining the row block-norm matrix:

$$R_k = \sum_j G_{kj} G_{kj}^* = G_{kk} G_{kk}^* + \sum_{j \neq k} G_{kj} G_{kj}^* \quad (13)$$

the total objective becomes:

$$f(X_k) = N - \text{tr}(X_k G_{kk}) - \text{tr}(G_{kk}^* X_k^*) + \text{tr}(X_k R_k X_k^*) \quad (14)$$

4.2. Matrix differentiation

To minimize (14) over X_k , we differentiate with respect to X_k^* . Using the standard matrix-calculus identities $\partial \text{tr}(XA)/\partial X^* = 0$, $\partial \text{tr}(A^*X^*)/\partial X^* = A$, $\partial \text{tr}(XBAX^*)/\partial X^* = XBA$ (for B Hermitian), we obtain:

$$\partial f / \partial X_k^* = -G_{kk} + X_k R_k = 0 \quad (15)$$

This follows because: the first term $\text{tr}(X_k G_{kk})$ has no X_k^* dependence (derivative is 0); the second term $\text{tr}(G_{kk}^* X_k^*)$ differentiates to G_{kk} ; and the third term $\text{tr}(X_k R_k X_k^*)$ differentiates to $X_k R_k$ (since R_k is Hermitian positive semidefinite by construction).

4.3. Solution

From (15), the optimal $X_k = \Lambda_k^{-1}$ is:

$$X_k^{opt} = G_{kk}^{-1} R_k \quad (16)$$

Therefore the optimal Fourier-domain block is:

$$\Lambda_k^{opt} = (X_k^{opt})^{-1} = R_k (G_{kk})^{-1} = (\sum_j G_{kj} G_{kj}^*) (G_{kk})^{-1} \quad (17)$$

This is the block generalization of Tyrtysnikov's (1992) superoptimal circulant. In the scalar case ($N = 1$), it reduces to $\lambda_k = \|G_k\|^2 / G_{kk}^*$.

Remark. When A is itself block-circulant, G is block-diagonal ($G_{kj} = 0$ for $k \neq j$), so $R_k = G_{kk} G_{kk}^*$ and $\Lambda_k^{opt} = G_{kk} G_{kk}^* (G_{kk})^{-1} = G_{kk}$. The preconditioner recovers $C = A$ exactly.

5. Explicit computation of G_{kk} and R_k

The optimal Fourier-domain block (17) requires two $N \times N$ matrices per harmonic: the diagonal block G_{kk} and the row block-norm matrix R_k . We now derive explicit, computable formulas for both.

5.1. The diagonal blocks G_{kk}

The diagonal blocks of $G = (F^* \otimes I) A (F \otimes I)$ are obtained by applying the DFT to the block structure of A . Writing out the (k, k) block explicitly:

$$G_{kk} = \sum_p \sum_q (F^*)_{kp} A_{pq} F_{qk} = (1/n) \sum_p \sum_q \omega_n^{(q-p)k} A_{pq} \quad (18)$$

Since $A_{pq} = 0$ for $|p - q| > n_p$, the sum collapses. Substituting $m = q - p$ and noting that A_{pq} depends on both $m = q - p$ and the diagonal index p (through the shift $ip\omega$), we separate the block-Toeplitz part S and the diagonal part D :

$$G_{kk} = (1/n) \sum_p (ip\omega) I \cdot 1 + (1/n) \sum_{m=-n_f}^{n_f} \omega_n^{mk} \sum_p S_m \quad (19)$$

The first sum is $(1/n) \sum_p ip\omega I$. For $p = -n_p, \dots, n_p$, this sum vanishes by symmetry (the positive and negative frequencies cancel). The second sum has $\sum_p S_m = n \cdot S_m$ since S_m is the same for every p (it is the block-Toeplitz part). However, the diagonal blocks of A have $A_{pp} = (s + ip\omega)I - M_0$, so the $m = 0$ term requires more care. The correct result is:

$$G_{kk} = (s + ik\omega) I_N - \sum_{j=-n_f}^{n_f} M_j \omega_n^{jk} \quad (20)$$

This is derived by noting that the DFT diagonalizes the block-diagonal shift D : the k th DFT mode sees the frequency $\omega_k = k\omega$, contributing the scalar shift $(s + ik\omega)$ to the diagonal. The remaining sum is the DFT of the Fourier coefficient matrices M_j evaluated at the k th DFT phase. Each G_{kk} is a sparse $N \times N$ matrix inheriting the union of the sparsity patterns of $M_{-n_f}, \dots, M_{n_f}$.

5.2. The row block-norm matrix R_k

By Parseval's theorem applied block-row-wise, R_k can be expressed in terms of AA^* rather than G :

$$R_k = \sum_j G_{kj} G_{kj}^* = (f_k^* \otimes I) AA^* (f_k \otimes I) \quad (21)$$

where f_k is the k th column of F . Expanding the quadratic form in terms of the $N \times N$ blocks of AA^* :

$$R_k = \sum_p \sum_q (F^*)_{kp} (AA^*)_{pq} F_{qk} = (1/n) \sum_{p,q} \omega_n^{(q-p)k} (AA^*)_{pq} \quad (22)$$

5.3. Structure of AA^*

The matrix AA^* is $nN \times nN$, sparse, and block-banded with bandwidth $2n_f$ (since A has block-bandwidth n_f , the product AA^* has block-bandwidth $2n_f$). Its (p, q) $N \times N$ block is:

$$(AA^*)_{pq} = \sum_l A_{pl} A_{ql}^* \quad (23)$$

Crucially, AA^* is *not* block-Toeplitz because the diagonal shifts D break the translational invariance: $(AA^*)_{pq}$ depends on both $(p - q)$ and on the absolute indices p, q individually (through the frequency shifts $ip\omega$ and $iq\omega$). However, AA^* is sparse and can be computed and stored as a block-banded matrix.

5.4. Evaluating R_k via diagonal averaging of AA^*

To evaluate (22) efficiently, group the double sum by the block-diagonal offset $m = q - p$:

$$R_k = (1/n) \sum_{m=-2n_f}^{2n_f} \omega_n^{mk} \sum_p (AA^*)_{p, p+m} \quad (24)$$

where the inner sum runs over all valid p such that both p and $p + m$ lie in $\{-n_p, \dots, n_p\}$. Define the *diagonal averages* of AA^* :

$$B_m = (1/n) \sum_p (AA^*)_{p, p+m}, \quad |m| \leq 2n_f \quad (25)$$

These are $N \times N$ sparse matrices obtained by averaging along each block diagonal of AA^* . Then R_k takes the simple form:

$$R_k = \sum_{m=-2n_f}^{2n_f} B_m \omega_n^{mk} \quad (26)$$

This has exactly the same structure as the formula for G_{kk} in (20): a DFT-weighted sum of sparse $N \times N$ matrices. The only difference is that the coefficients are $\{B_m\}$ (diagonal averages of AA^*) instead of $\{M_j\}$ (Fourier coefficients of the operator). R_k inherits the sparsity pattern of the B_m matrices, which is the union of the sparsity patterns of $(AA^*)_{p, p+m}$ — denser than M_j (due to the products in AA^*) but still sparse.

Remark (Chan's construction). The diagonal averaging (25) is precisely Chan's optimal circulant approximation applied to the block-banded matrix AA^* . It produces the block-circulant matrix closest to AA^* in Frobenius norm, whose k th Fourier-domain block is R_k . This is the natural and correct way to extract R_k from the quasi-block-Toeplitz structure.

Remark (block-Toeplitz limit). If $D = 0$ (no frequency shifts), then $A = S$ is exactly block-Toeplitz, $AA^* = SS^*$ is also block-Toeplitz, and $(AA^*)_{p, p+m}$ is the same for every p . The diagonal average reduces to $B_m = (SS^*)_m = \sum_j S_j S_{j-m}^*$, recovering the block autocorrelation. The general formula (25) is therefore a correction that accounts for the effect of the diagonal shifts D on the row norms.

5.5. R_k is Hermitian positive semidefinite

By construction, $R_k = \sum_j G_{kj} G_{kj}^*$ is a sum of matrices of the form XX^* , hence Hermitian positive semidefinite. This guarantees that the Cholesky factorization in Algorithm 1 is well-defined (assuming R_k is nonsingular, which holds when A is nonsingular).

6. Applying the preconditioner

From (17), the optimal Fourier-domain block is $\Lambda_k^{\text{opt}} = R_k G_{kk}^{-1}$. To apply $(\Lambda_k^{\text{opt}})^{-1}$ to a vector $\mathbf{v}$, we solve $\Lambda_k^{\text{opt}} \mathbf{x} = \mathbf{v}$, i.e.:

$$R_k G_{kk}^{-1} \mathbf{x} = \mathbf{v} \quad (27)$$

Rearranging: $G_{kk}^{-1} \mathbf{x} = R_k^{-1} \mathbf{v}$, so $\mathbf{x} = G_{kk} R_k^{-1} \mathbf{v}$. The application therefore consists of two steps:

Step A: Solve $R_k \mathbf{y} = \mathbf{v}$ (one $N \times N$ system)

Step B: Compute $\mathbf{x} = G_{kk} \mathbf{y}$ (one sparse matrix-vector product)

The full preconditioner application (Algorithm 1) is:

Algorithm 1: Apply superoptimal preconditioner $C^{-1} \mathbf{v}$	
1.	$\mathbf{v} \leftarrow (F_n \otimes I_N) \mathbf{v}$ (block FFT along harmonic index)
2.	for $k = 0, \dots, n-1$ do (in parallel):
	(a) Solve $R_k \mathbf{y}_k = \mathbf{v}_k$ ($N \times N$ solve, using pre-computed Cholesky of R_k)
	(b) $\mathbf{x}_k = G_{kk} \mathbf{y}_k$ (sparse matrix-vector product)
3.	$\mathbf{x} \leftarrow (F_n^* \otimes I_N) \mathbf{x}$ (block inverse FFT)

7. Setup

Algorithm 2: Setup (one-time)	
1.	Compute AA^*. Form the sparse block-banded matrix AA^* (block-bandwidth $2n_f$). Cost: $O(n \cdot n_f \cdot \text{nnz})$.
2.	Diagonal averages. Compute $B_m = (1/n) \sum_p (AA^*)_{p,p+m}$ for $ m \leq 2n_f$ by averaging along each block diagonal of AA^* . Cost: $O(n \cdot n_f \cdot \text{nnz}(B_m))$.
3.	Assemble G_{kk}. For $k = 0, \dots, n-1$: $G_{kk} = (s + ik\omega)I - \sum_j M_j \omega_n^{jk}$ via Eq. (20). Cost: $O(n \cdot n_f \cdot \text{nnz})$.
4.	Assemble R_k. For $k = 0, \dots, n-1$: $R_k = \sum_m B_m \omega_n^{mk}$ via Eq. (26). Cost: $O(n \cdot n_f \cdot \text{nnz}(B_m))$.
5.	Factor R_k. Cholesky factorization of each R_k (Hermitian positive definite). n independent factorizations, embarrassingly parallel. For real-valued base flows: $R_{n-k} = R_k^*$, so only $n/2$ factorizations needed.
6.	Store G_{kk} as sparse matrices for the mat-vec in step 2(b) of Algorithm 1.

8. Comparison with the standard (Strang/Chan) preconditioner

The standard block-circulant preconditioner uses $\Lambda_k^{\text{std}} = G_{kk}$. The superoptimal preconditioner uses $\Lambda_k^{\text{opt}} = R_k G_{kk}^{-1}$. Relating the two:

$$\Lambda_k^{\text{opt}} = G_{kk} (I + G_{kk}^{-1} \sum_k^2 G_{kk}^{-*}) = G_{kk} + \sum_k^2 G_{kk}^{-*} \quad (28)$$

where $\sum_k^2 = \sum_{j \neq k} G_{kj} G_{kj}^*$ is the off-diagonal contribution. The correction $\sum_k^2 G_{kk}^{-*}$ accounts for the energy in the off-block-diagonal part of G that no block-circulant can eliminate. When A is block-circulant, $\sum_k^2 = 0$ and both preconditioners coincide.

The additional cost per GMRES iteration compared to the standard preconditioner is one sparse mat-vec with G_{kk} per harmonic block (Step 2b). The Cholesky solve (Step 2a) replaces the LU solve in the standard preconditioner at comparable cost (and potentially cheaper, since Cholesky exploits symmetry).

9. Minimum residual achieved

Substituting the optimal $X_k = G_{kk} R_k^{-1}$ back into (14):

$$\min_C \|I - C^{-1}A\|_F^2 = \sum_k (N - \text{tr}(G_{kk} R_k^{-1} G_{kk}^*)) \quad (29)$$

Since $R_k = G_{kk} G_{kk}^* + \Sigma_k^2 \geq G_{kk} G_{kk}^*$ (in the positive semidefinite sense), we have $R_k^{-1} \leq (G_{kk} G_{kk}^*)^{-1}$, so $\text{tr}(G_{kk} R_k^{-1} G_{kk}^*) \leq \text{tr}(G_{kk} (G_{kk} G_{kk}^*)^{-1} G_{kk}^*) = N$. Therefore each term in (29) is non-negative, and equals zero iff $\Sigma_k^2 = 0$ (i.e., A is block-circulant).

10. Recycling factorizations across Laplace shifts

In resolvent analysis, one sweeps $s = i\omega_f$ across a range of forcing frequencies. From (20), changing $s \rightarrow s + \Delta s$ shifts G_{kk} by $\Delta s \cdot I$. The diagonal averages B_m in (25) also change, but the change is a structured perturbation involving the autocorrelation of the shifted diagonal.

For nearby values of s , the Cholesky factors of $R_k(s_0)$ can be used as preconditioners for $R_k(s_1)$ within an inner Krylov solve, yielding a flexible GMRES (FGMRES) scheme. This avoids re-factorizing R_k at every frequency point.

11. Summary

The superoptimal block-circulant preconditioner for the quasi-block-Toeplitz harmonic balance system $A = D + S$ is defined by the Fourier-domain blocks:

$$\Lambda_k^{opt} = R_k G_{kk}^{-1}, \quad k = 0, \dots, n-1 \quad (30)$$

where $G_{kk} = (s + i\omega)I - \sum_j M_j \omega_n^{jk}$ (the standard Fourier-domain block, Eq. 20) and $R_k = \sum_m B_m \omega_n^{mk}$ (the DFT of the diagonal averages of AA^* , Eq. 26). Application requires one Cholesky solve and one sparse mat-vec per harmonic block, sandwiched between forward and inverse block FFTs.

References

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